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Options and Futures

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Fall Term 2025

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The FTAP and Risk-Neutral Pricing in Continuous Time

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- Dynamic version of the FTAP
- Risk-neutral pricing in continuous time
- Itô processes under Q
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Dynamic version of the FTAP

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- There are N risky assets. For $n = 1, \ldots, N$, the price of asset n is given by an Itô process π_n.
- In addition, there is a money market account (MMA) with an instantaneously risk-free interest rate r_t.
- r_t itself can follow a stochastic process.
- If the position in the MMA is B_t at time t, then

$$dB_t = r_t B_t dt,$$

or equivalently

$$B_t = e^{\int_0^t r_u du} B_0.$$

The FTAP and risk-neutral valuation

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- **FTAP:** There is no arbitrage if and only if there exists a probability measure Q such that for any non-dividend paying asset, whose value at time t is given by π_t

$$\pi_t = \mathbb{E}_t^Q \left[e^{-\int_t^T r_u du} \pi_T \right],$$

for any $T > t$[1]

- Under Q, the price of a security (or the value of any trading strategy or derivative) is equal to the expectation of its value at any future point in time discounted at the risk free rate.
  - For this reason Q is referred to as a risk-neutral probability measure.

[1] This is an imprecise statement. Existence of Q implies no arbitrage, but only a stronger version of no arbitrage implies existence of Q

Risk-neutral pricing (cont'd)

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- For the case of a constant riskfree rate r (as in BS), the risk-neutral valuation formula

$$\pi_t = \mathbb{E}_t^Q \left[e^{-\int_t^T r_u du} \pi_T \right]$$

reduces to

$$\pi_t = e^{-r(T-t)} \mathbb{E}_t^Q(\pi_T).$$

Itô processes under Q

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- In order to evaluate an expectation under Q, we need to know the distribution of the price process under Q.
- Denote a standard Brownian motion under the actual measure P by W, and a standard Brownian motion under the risk-neutral measure Q by W^Q.
- Since π is an Ito process under P we can write it as follows:

$$d\pi_t = \mu_t \pi_t dt + \sigma_t \pi_t dW_t.$$

- **Fact:** Under Q, π is an Itô process with a different drift but the same diffusion. For a non-dividend paying asset, the instantaneous expected rate of return changes from μ_t to r_t

$$d\pi_t = r_t \pi_t dt + \sigma_t \pi_t dW_t^Q.$$

Dividend-paying asset

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- Consider a dividend-paying asset, with ex-dividend price process under P given by
$$d\pi_t = \mu_t \pi_t dt + \sigma_t \pi_t dW_t.$$
- Suppose the dividend yield is given by a stochastic process q
$$dD_t = q_t \pi_t dt.$$
- Under Q, the ex-dividend price process becomes
$$d\pi_t = (r_t - q_t)\pi_t dt + \sigma_t \pi_t dW_t^Q.$$
- Under Q, the cum-dividend price process is
$$d\pi_t + dD_t = r_t \pi_t dt + \sigma_t \pi_t dW_t^Q.$$
- Under Q, the total instantaneous expected return on any asset at time t isequal to r_t

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Black-Scholes redone

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- There are three assets:
  - A stock: $dS_t = \mu S_t dt + \sigma S_t dW_t$,
  - A MMA with constant instantaneous interest rate r,
  - A derivative on the stock with payoff $g(S_T)$ at time T.
- FTAP $\Rightarrow$ there is a probability measure Q such that
  - The time t price of the derivative f_t is given by

$$f_t = e^{-r(T-t)}\mathbb{E}_t^Q[g(S_T)];$$

  - Under Q, S has drift r; the diffusion remains the same as under P:

$$dS_t = rS_t dt + \sigma S_t dW_t^Q$$

$$\Rightarrow \log S_T \sim \mathcal{N}[\log S_t + (r - \sigma^2/2)(T-t), \sigma^2(T-t)]$$

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Black-Scholes redone (cont'd)

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- This is exactly the probabilistic representation of the derivative price function which we obtained by applying the Feynman-Kac formula.
- In particular, we know how to compute

$$e^{-r(T-t)}\mathbb{E}_t^Q[(S_T - K)^+].$$

This leads to the Black-Scholes formula for a European call option.

- If the underlying pays a constant dividend yield q , we have

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t^Q.$$

- Check that the above risk-neutral valuation formula gives us the BS formula for an underlying with a constant dividend yield.

The discrete-time case

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- Recall that the risk-neutral valuation formula in the continuous-time case is
$$\pi_t = \mathbb{E}_t^Q \left[e^{-\int_t^T r_u du} \pi_T \right].$$
- Now consider the discrete-time case, with trading at dates $0, 1, 2, \ldots, T$[2]
- The interest rate is r_t from t to $t+1$, with continuous compounding, and follows an arbitrary stochastic process.
- Then the risk-neutral valuation formula is
$$\begin{aligned} \pi_t &= \mathbb{E}_t^Q \left[e^{-r_t} e^{-r_{t+1}} \ldots e^{-r_{T-2}} e^{-r_{T-1}} \pi_T \right] \\ &= \mathbb{E}_t^Q \left[e^{-\sum_{s=t}^{T-1} r_s} \pi_T \right]. \end{aligned}$$

[2]Notice that this implies that the time unit is chosen to be the time interval between successive trading dates; e.g. we measure time in days if trading is at daily intervals.

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- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

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- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

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Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

Breeden-Litzenberger

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- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

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Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

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- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

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